FETCH

The update of 11 October 2026

What was fixed after the QA test, what the forecast does now, and what is still open

This is the plain-language account of the work done after the QA test report. It says what was wrong, what happens now, how each change was checked, what was not fixed and why, and what is waiting for a decision.

Everything described here is on the work branch (pro-redesign). Nothing has been deployed: the live site is exactly as it was. Nothing was written to the shared database.

The row-by-row record is QA/QA-TEST-REPORT.md. The forecast is described in QA/FORECAST-METHOD.md.

1. In short

The QA test	96 test cases. 63 passed, 27 failed, 6 could not be run without saving real records.
After this update	Of the 27 that failed: **21 are fixed**, **2 are partly fixed**, and **4 are left open** for a stated reason.
The forecast	It ran 36.4% above what sold. It now runs **15.6%** above (18.9% in the strictest test). The average miss per product went from 1.55 items to **1.40**.
The live site	Unchanged. None of this is on it until it is deployed.
The shop's data	Unchanged. Every fix that saves anything was tried on a separate test database on this computer. The 199 sales without items were not touched.
Still needed from you	A yes or no on deploying; a decision on six test cases that need real saves; a decision on the sales data; and changes to the paper, slides and test plan so they describe the same method as the system.

How each fix was checked

- **On a separate test database.** A private copy of the database structure, holding only TEST_ records, on this computer. The real server code ran against it, so a fix was tried by really saving, not by reading the code.
- **In a test browser**, wherever the fix shows on a screen: the button was pressed, the message read, and the requests counted.
- **Read-only on the real sales**, for the forecast: the real history was read and nothing was written.
- **By automated tests.** 184 of them now, all passing.
- **By a second review.** After the report was updated, each of its claims was checked against the code by separate reviewers. They found no fix that was missing or did not do what was said. They corrected details in 17 places, moved one case from "fixed" to "partly fixed", and found one small fault, which is now fixed. That last fix is the one exception to the first point: it was checked by reading the code and by the automated tests only.

> **"Fixed – retest pending"** is how fixed cases are marked in the report. It means: fixed and checked as above, but not yet tried on the real system. That retest can only happen after deploying.

2. What was fixed

Grouped by what a user would notice. The code in the last column is the commit that made the change.

What a panelist sees in the first five minutes

What was wrong	What happens now	Commit
The Dashboard showed every sale as "Completed", voided ones included	Each sale shows its own status, and a voided sale is marked	acde06b
The Dashboard counted voided sales as sales (2,341 against 2,338 in Reports), and its totals took 2.5 seconds	Voided sales are left out and shown as "N more voided". The totals take about 0.1 to 0.2 seconds	bfd77f9
"Low stock" was 6 on one card and 4 on the chip beside it; "Overstock" meant two different things	The two cards are renamed "Restock needed" and "Over max level", so one word means one number	8dd5f9f
Revenue by category did not add up to revenue (₱8,847,885 against ₱9,983,522.50)	Sales recorded without items and discounts are shown as their own lines, and the list adds up to the centavo	3c0e0a1

Saving safely

What was wrong	What happens now	Commit
Double-clicking "Save adjustment" changed the stock twice. A second Charge could go through while the first sale was still saving	The button is locked while saving, and the server saves a repeated request only once	93ba109
Removing 999,999 from a product with 20 in stock was accepted without a word	Refused: "Only 20 in stock: you cannot remove 999,999." Checked on the page and again on the server	9051de7
The same product on two lines of one sale could sell more than the stock	Lines for one product are added together before the stock check	8cf9ba4
Suppliers could be added twice, with a name of only spaces, and with any text as email or phone	A name already in use is refused; the name, email and phone are checked	794c500
A product that does not exist was "deleted successfully". A product with stock was deleted with no warning. Two products could share a name by accident	A missing product is answered "not found". The question names the product and its stock. A name already in use is asked about first	c535a0a
A price of 1,000,000,000,000, a 300-letter name or letters as a quantity passed the server's checks	Each is refused with the field named	278e6ff 2547ac2

What was wrong	What happens now	Commit
A list such as [3] sent as a quantity was read as 3	Refused: a quantity has to be a number	ce584e3
An edit that was refused had already uploaded its picture	The picture is stored last, after every check	90bed5f

Who can do what

What was wrong	What happens now	Commit
A view-only account could mark every notice read and clear them for everyone	Only an administrator or a manager can	352845e
A manager could delete a sale for good, and the Activity log kept no copy of it. Deleting a voided sale returned its stock a second time	Only an administrator can delete. The log keeps the sale's number, date, cashier, customer, totals and every item. Stock is returned once	217dfba
Sign-in answered an unknown user name in 2 ms and a wrong password in 118 ms, so anyone could find out which names exist	Both now take the same time (about 121 ms) and get the same answer	ce07f9b
Every answer from the server announced "X-Powered-By: Express"	No longer sent	7698cb6

Wording and small things

What was wrong	What happens now	Commit
In the stock window, "Adjustment" and "Damage" both took stock away, and nothing said so	The four choices each say which way they go, and the window shows the stock before and after	2547ac2
The Users screen said "Delete user" for what is really a deactivation	It says "Deactivate", names the account, and says it can be restored	ce584e3
Typing % or _ in a search matched every row	They are looked for as the characters they are	d07958e
A report asked for with dates that do not exist returned the whole history with no message	Refused with the reason: "The start date is not a real date…", "The start date is after the end date."	d07958e
Between midnight and 8 am the Sales export was named with yesterday's date. The backup was a zip saved under a .json.gz name	Both carry the shop's date, and the backup is saved as .zip	4fbd723

The forecast

What was wrong	What happens now	Commit
One freak day in one product (say 10,000) raised every other product's forecast by 130%	A day counts for no more than a cap: the larger of 5, and 4 times the product's typical selling day. The same test now adds 0.3 of a percentage point	270ae10
A new product that sold 5 on its first day was forecast at 150 for the month, with an order suggested for 165	Until it has 14 days on record its sales are spread over 14 days: 10.8 for the month, no order	54a6680
22 days in May, when sales were recorded without items, were read as days when nothing sold	Those days (10 to 31 May) are left out of every product's history. Nothing in the database is changed	fbc407d
The forecast ran 24% to 43% high, and the machine-learning part did worse than the simple trend	Settings re-chosen by a test on fortnights the forecast had not seen, plus a last step that corrects half of the most recent miss. See section 3	3f4b61e
14 products were marked "high" confidence. They were exactly the 14 that had stopped selling	No sales is no longer scored as "perfectly steady sales". All products now read "low"	377a063
The page did not say what the forecast cannot know	A section "What the forecast cannot know" is on the Stock planning page	d6871c1
The help guide and the assistant described the method differently from the page	All three now describe what the code does	5fb2f31

3. The forecast, before and after

Tested on four fortnights it was not shown (16 August to 10 October 2026): 120 products in each, 780 items sold in all.

	Before	Now
Forecast total against 780 sold	1,064 (+36.4%)	**901 (+15.6%)**
Average miss per product per fortnight (MAE)	1.55 items	**1.40 items**
Large misses (RMSE)	1.92	**1.74**
Better than the best rule of thumb by	2%	**11%**

Fortnight	Sold	Shop-total accuracy before	Now
16 to 29 August	235	63.5%	70.0%
30 August to 12 September	210	57.3%	78.7%
13 to 26 September	175	75.8%	90.9%
27 September to 10 October	160	58.5%	86.1%

Product group	Ran high by, before	Now
Food	+67.9%	+37.2%
Health & Wellness	+18.8%	+2.9%
Accessories	+16.1%	+2.8%
Litter & Waste	+37.6%	+15.8%

What changed in the settings

Setting	Before	Now
How far back the model reads	365 days	120 days
How older days count	Every day the same	A day's weight halves every 60 days
The model's share of the final figure	Half	All of it, for a product with 6 weeks of sales
Last step	None	Corrected by half of its own last miss, by category

> **Which figure to quote.** The settings were chosen by scoring each fortnight with settings picked only from earlier fortnights. That strict test gives **+18.9% and an average miss of 1.45 items**. Quote that

> one. The +15.6% is the system as it now stands, on fortnights that also helped choose its final settings, so it is slightly flattering.

The three figures that were on the Stock planning screen (35.2%, 82.2% and 58.5%) were accuracy for the shop total in one fortnight, not error. The same test now gives 86.1% for the latest fortnight.

4. What was not fixed, and why

What	Why it is as it is
Two cases are only partly fixed: the forecast still runs high (B04), and the two models alone are still no closer to the shop total than the simple trend (B06)	The forecast is above what sold in three of the four fortnights. Per product the models now have the smaller miss, and the figure the page uses is closer to the total in all four
A product in sharp decline is kept too high (B12, L4)	A rule for "a clear drop in level" was tried. On the four test fortnights it never acted on a single real product, so it could not be tested on this shop's data. It also carries a risk: an empty shelf looks the same as a drop in demand. It was not added
A lasting rise is recognised slowly	This is new, and it is the price of the new settings. A made-up product that went from 1 sale every 5 days to 1 a day was only half recognised after about six weeks. The settings were chosen in months when sales only fell
The sales data: 199 sales with no items, cost equal to price for 119 of 120 products, 59 stock movements for 4,473 sale lines (C04, C05, L13)	Left untouched on your instruction. Their effects on the forecast and on the reports were fixed; the data itself is yours to decide
The method named in the test plan (B02, H4)	The system now says one thing in every place. The test plan still says linear regression; the documents are yours to change
The security policy still allows scripts written into the page (half of L9)	The pages carry about 287 such handlers. It cannot be tightened without rewriting them

New things to be ready for

- Every product now reads **"low" confidence**. That is the honest reading: no product sold on more than 13 of the last 56 days.
- A product whose sales rise for good will be **under-forecast for some weeks**. Check fast-rising products by eye.
- The forecast's settings were chosen in a period when **sales only fell**. Run the tuning test again when a few more months of sales exist.

5. Waiting for your decision

Deploying

Nothing is deployed. Deploying means pushing the work branch to main, which the hosting picks up by itself. After that, the cases marked "Fixed – retest pending" can be tried on the real system.

Your own server on this computer was started before these fixes and has since stopped. Starting it again runs the start-up steps and the hourly checks against the shared database, so that is yours to do or to approve.

Six test cases that need real saves

None was run. The rows below were counted by doing the same actions on the test database.

Case	What it would create in the shared database	How it would be removed
A TEST_ product, added, edited and deleted	Adding: 1 product, 1 log row, 1 notice. Editing: 1 log row. Deleting in the app: 1 log row; the product row stays, hidden	The app never erases a product. Erasing the hidden row needs a direct delete in the database, which is a separate yes
Stock in and stock out on it	Each change: 1 stock history row, 1 log row, 1 notice	As above
A real sale at the till	1 sale, 1 line per product, 1 stock history row per line, 1 log row, 1 notice (2 if it takes the product to its reorder level). If the cashier has not opened the till today: 1 till-opening row and 1 log row. Revenue goes up until the sale is voided	Void it (the sale stays, marked voided, out of every total), then delete it as administrator (the log keeps a copy)
An alert appearing and clearing	Going below the reorder level: 1 stock history row, 1 log row, 2 notices. Restocking: the same with 1 notice	As above. "Mark all read" and "Clear read" would not be pressed: they act on everyone's notices
Page speed on the live site, signed in	One sign-in: 1 log row, 1 notice. Opening pages creates nothing, except the Notifications page, which makes the server run its hourly stock check at once	Nothing but those two rows
A real sign-in, sign-out and session expiry	Each sign-in: 1 log row, 1 notice. Signing out creates nothing. Expiry can only be seen by waiting 24 hours	As above
Also part of the first case: an end-of-day count	1 count row and 1 log row. That cashier can then sell nothing more that day	A manager reopens the count: the count row is deleted, 1 more log row

Case	What it would create in the shared database	How it would be removed
Also part of the first case: receiving an order	A purchase order first: 1 order with its lines, 1 log row, 1 notice and an email to the supplier. Each later step: 1 log row and 1 notice. Receiving adds the stock	The app cannot delete a purchase order; it would stay, marked received

- **Email.** Suppliers of low-stock products are emailed at most once a day. For the alert case the TEST_ product would be given a TEST_ supplier whose address is your own, so no real supplier is written to.
- **Who signs in.** Passwords are not typed for you. Either you sign in and the result is watched, or you create temporary accounts, deactivated afterwards.
- **What "removed" means.** Voiding and deleting in the app leave the Activity log rows and the notices, by design. Removing those too means deleting rows from the database directly.
- **"Mark all read" and "Clear read"** would not be pressed in any of these: they act on the shared notices that everyone sees.

The documents

QA/FORECAST-METHOD.md, section 12, lists what to change in the paper, the slides and the test plan so that all three describe the method the system really uses.

The sales data

The 199 sales without items, the product costs and the stock history are unchanged. Whatever is decided about them, the paper should say how the sales history was made, because every accuracy figure describes the method on this history.

6. Every change, in order

The newest is last. Each line is one commit on the work branch.

Commit	What it did
96d41f0	The QA test report, as first written
acde06b	Dashboard: each sale shows its real status
93ba109	One press saves once: stock changes, and a sale at the till
9051de7	More stock cannot be removed than there is
352845e	A view-only account can no longer mark notices read or clear them
270ae10	Forecast: a freak day is capped
54a6680	Forecast: a new product gets a cautious first estimate
fbc407d	Forecast: days with sales but no items are left out
3c0e0a1	Reports: revenue by category adds up to revenue
3f4b61e	Forecast: settings re-chosen on held-out fortnights; corrected for its own last miss
794c500	Suppliers: no duplicates; details checked
8cf9ba4	A sale: lines for one product are added together before the stock check
217dfba	Deleting a sale: administrator only, with a copy kept
e1bfb46	The tuning test, as a script that can be run again
d6871c1	Stock planning: the page says what the forecast cannot know
bfd77f9	Dashboard totals: fast, and voided sales not counted
8dd5f9f	Inventory: two cards renamed
2547ac2	Stock window: each choice says which way it goes
278e6ff	Products: oversized values refused with the field named
c535a0a	Products: deleting says what it does; a name already in use is asked about
d07958e	Search and report dates
4fbd723	Download names; the backup saved as a zip
7698cb6	The server no longer announces its framework
ce584e3	Wording: deactivate, not delete; a list is not a number
377a063	Forecast: confidence no longer "high" for products that stopped selling
ce07f9b	Sign-in: the answer no longer shows whether a user name exists
9f72d40	QA/FORECAST-METHOD.md written

Commit	What it did
5fb2f31	Help guide and assistant describe the forecast as it now works
f0386f8	The plain-language guide (Word)
90bed5f	Products: a refused edit no longer uploads its picture first

Documents in the QA folder

File	What it is
QA-TEST-REPORT.md	The test report, with every fixed case marked and its commit
FORECAST-METHOD.md	How the forecast works now, with its test results and how to repeat them
FETCH-System-and-Analytics-Guide.docx	Each function of the system and the analytics, in plain words
FETCH-Analytics-Explanation-and-Computation.docx	The analytics with every computation worked through
FETCH-Update-11-October-2026.docx	This document